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Nuclear Electric Propulsion for Deep-Space Missions: Technologies, Challenges, and Future Outlook

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Abstract — Nuclear electric propulsion (NEP) is increasingly recognized as a cornerstone technology for future deep-space missions, offering solutions to the performance limits of conventional chemical propulsion. This review consolidates recent progress in NEP system architectures, with particular focus on compact fission reactors, high-efficiency power conversion units, and advanced electric thrusters, such as ion engines, Hall-effect devices, and magnetoplasmadynamic systems. The technological trajectory is traced from early demonstrations, including SNAP-10A and the Soviet TOPAZ reactors, to current initiatives such as the U.S. National Aeronautics and Space Administration's kilopower and DRACO projects, the European Space Agency's Future Launchers Preparatory Programme, and Roscosmos's Transport and Energy Module.

Critical challenges, ranging from reactor miniaturization and thermal management at high operating temperatures to radiation protection and regulatory approval, are examined alongside emerging solutions such as artificial intelligence–assisted autonomy, digital-twin modeling, and metamaterial radiator concepts for megawatt-class platforms.

To structure this landscape, we introduce a mission taxonomy based on reactor output [<10 kW(electric), 10 to 100 kW(electric) > 100 kW(electric)] and provide new analyses of radiator mass scaling as well as comparative performance of NEP and alternative propulsion systems for Mars logistics. Looking forward, pathways include modular reactor arrays, hybrid propulsion architectures combining NEP with solar-electric or chemical systems, and the establishment of coordinated international policy frameworks. Together, these elements outline a roadmap for advancing NEP from experimental demonstrations to operational systems, reinforcing its role as a foundation for sustainable interplanetary exploration.

Keywords — Nuclear electric propulsion, electric thrusters, space reactors, deep-space logistics, artificial intelligence.

I. INTRODUCTION

As crewed and robotic missions extend beyond low Earth orbit, the constraints of conventional chemical propulsion become more apparent. Although chemical launch vehicles have been the backbone of current spaceflight, their relatively modest specific impulse (Isp) (~ 450 s) and heavy propellant requirements limit their suitability for long-duration, high-energy deep-space trajectories [1,2]. To overcome these limitations,

increasing attention has turned to nuclear propulsion technologies, which can be broadly divided into nuclear thermal propulsion (NTP) and nuclear electric propulsion (NEP).

In NTP, reactor heat is used directly to accelerate the propellant, providing high thrust but only moderate Isp. In contrast, NEP decouples power generation from thrust production, allowing for the use of high-efficiency electric thrusters with Isp's in the range of 3000 to 10 000 s, well suited for the continuous low-thrust maneuvers required in interplanetary missions [1,3,4]. These characteristics make NEP especially attractive for applications such as robotic cargo

delivery, sustained planetary logistics, and eventual human expeditions to Mars [5].

Despite decades of conceptual and experimental progress, comprehensive reviews of NEP remain limited. Much of the existing literature addresses only subsystems or isolated mission studies [6,7] without integrating aspects such as reactor design, power conversion approaches, electric thruster technologies, and thermal control solutions within a broader international context.

This review seeks to fill that gap by providing a systematic assessment of NEP technologies and the engineering challenges that accompany them.

It revisits early milestones, including the U.S. Systems for Nuclear Auxiliary Power (SNAP)-10A and the Soviet TOPAZ series reactors [8,9], highlights ongoing programs such as the U.S. National Aeronautics and Space Administration's (NASA's) Kilopower and Demonstration Rocket for Agile Cislunar Operations (DRACO) projects [10,11], and considers emerging directions in autonomy, modular spacecraft design, and artificial intelligence (AI)-driven mission control. Together, these perspectives aim to guide both researchers and mission architects in advancing NEP systems from research prototypes toward operational deployment. A conceptual comparison of NEP and NTP architectures is presented in Fig. 1.

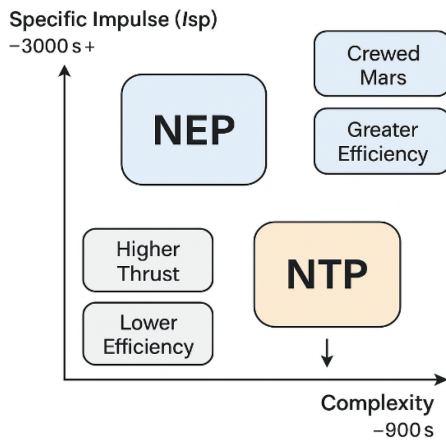


Fig. 1. Conceptual comparison of NEP and NTP systems in terms of Isp, thrust, efficiency, and system complexity. NEP systems provide significantly higher Isp (~3000 s) and greater efficiency, making them suitable for long-duration missions such as cargo or deep-space exploration. In contrast, NTP systems offer higher thrust and simpler architectures with lower Isp (~900 s), making them favorable for crewed interplanetary missions such as Mars transit.

I.A. Review Methodology

This review employed a narrative synthesis framework to examine the technological, operational, and regulatory dimensions of NEP. The survey drew on a wide range of sources, including peer-reviewed journal publications, technical reports from government agencies, NASA archival documents, and proceedings from international propulsion conferences. Key databases consulted were the NASA Astrophysics Data System, IEEE Xplore, ScienceDirect, and Google Scholar. The literature span covered 1960 through 2025, with particular emphasis on studies published after 2000 to capture contemporary developments.

Priority was placed on works featuring experimental validation, prototype demonstrations, and official technology roadmaps sponsored by national space programs. The figures and tables were either adapted from established references or generated by the author through consolidation of available data. A comparative overview of major propulsion technologies is presented in Table 1.

II. FUNDAMENTALS OF NEP

II.A. Working Principles

NEP systems separate power generation from thrust production, enabling continuous operation, with very high Isp values an essential capability for long-duration deep-space missions. At the core of a NEP system is a compact nuclear fission reactor that produces thermal energy through carefully controlled chain reactions [3,4]. This thermal output is then converted into electricity using advanced conversion technologies such as Brayton-cycle turbines, thermoelectric modules, or thermionic converters, each offering distinct trade-offs in terms of efficiency, scalability, and reliability [1–6].

The electrical power generated is subsequently supplied to electric thrusters, where propellants, which are most often inert gases such as xenon, are ionized and accelerated by electrostatic or electromagnetic fields to generate thrust [7,8]. With this arrangement, NEP systems routinely achieve Isp levels in the range of 3000 to 10 000 s, far surpassing those of chemical propulsion [9,10].

While the thrust-to-weight ratios are comparatively modest, they remain adequate for interplanetary missions. Overall efficiency is governed by the combined effectiveness of converting reactor thermal energy into electricity,

TABLE 1
Verified Comparison of Major Propulsion Technologies [1,2,4,6,9–20]

Parameter	Chemical Propulsion (Liquid Hydrogen/Liquid Oxygen)	Ion Propulsion	HET	NEP (with Ion/HETs)	VASIMR
Isp	~450 s (liquid hydrogen/ liquid oxygen, e.g., Space Shuttle Main Engine)	3000 to 10 000 s typical	1500 to 2500 s (commercial); up to 3000 s experimental	Same as ion/Hall variants noted previously	2000 to 5000 s demonstrated; theoretical up to ~10 000 s
Thrust	10^5 to 10^6 N	~0.1 to 0.3 N (high- power ion)	0.04 to 0.6 N typical; laboratory up to ~5 N	Same as ion/Hall (depends on power source)	0.1 to 5 N scalable
Efficiency	~55% overall thermal efficiency	~60% to 80%	~45% to 60%, up to ~75%	~60% to 85% thruster system efficiency	≥70% conversion to jet power
Power source	Chemical	Solar/nuclear	Solar/nuclear	Nuclear reactor feeds electric thrusters	Solar/nuclear
Operational duration	Minutes (burn time limited)	Months to years	Months to years	Years	Months to years
Complexity	Low	High	Medium	Very high	High
Technology readiness	TRL 9 (flight proven)	TRL 7 to 9	TRL 7 to 9	TRL 4 to 6 (megawatt- scale NEP emerging)	TRL 4 to 6 (laboratory grade)

and subsequently transforming that electrical power into propulsive kinetic energy [11].

II.B. Key Components

II.B.1. Nuclear Reactor Core

NEP systems rely on compact, space-optimized fission reactors as their primary energy source. These reactors vary in design and technology readiness levels (TRLs):

1. *Solid-core reactors*: Mature technology, exemplified by NASA's Kilopower project, which uses ^{235}U fuel and Stirling engines for power conversion [12].

2. *Gas-core reactors*: Capable of higher operating temperatures and power densities, though still at the conceptual stage [13].

3. *Thermionic reactors*: As demonstrated in the Soviet TOPAZ series, these systems convert heat directly to electricity via thermionic emission [21].

4. *Fission-fragment reactors*: Advanced designs that use charged fission fragments for either direct propulsion or high-efficiency power generation [14].

II.B.2. Power Conversion Units

The efficiency of NEP systems is largely dependent on the performance of the power conversion technologies:

1. *Thermoelectric systems*: Mechanically simple with no moving parts, but limited by low conversion efficiencies (~5% to 7%) [22].

2. *Brayton cycle turbines*: Employ gas expansion to drive generators, achieving higher efficiencies (~25% to 30%) suitable for megawatt-class applications [23].

3. *Thermionic converters*: Compact and moderately efficient (~10% to 15%), directly converting heat into electricity [15,21].

II.B.3. Electric Thrusters

Electric thrusters serve as the final stage in the NEP chain, providing sustained propulsion:

1. *Ion thrusters*: Used in missions like NASA's Deep Space 1 and Dawn, offering high Isp's and proven operational heritage [16,24].

2. *Hall-effect thrusters (HETs)*: Widely deployed in Russian and European satellites due to their robustness and moderate power requirements [25].

3. *Magnetoplasmadynamic (MPD) thrusters*: Capable of high thrust densities for high-power systems, but face efficiency and lifetime challenges [26].

4. *Variable Specific Impulse Magnetoplasma Rocket (VASIMR)*: Enables adjustable Isp-thrust trade-offs, currently under testing for high-power missions [27].

Fig. 2 shows a schematic architecture of a basic NEP spacecraft. In summary, the successful operation of NEP systems hinges on the seamless integration of high-efficiency reactors, reliable power conversion units, and adaptable electric thrusters. Each subsystem must operate harmoniously under the constraints of space, including microgravity, radiation exposure, and thermal vacuum conditions. Optimizing the interface between these units

is essential for enabling scalable, autonomous propulsion platforms for next-generation interplanetary missions.

II.B.4. Comparative Analysis of Propulsion Systems

Table 2 presents a comparative analysis of representative propulsion systems.

This comparative analysis highlights NEP's advantages in Isp and operational duration, positioning it as an ideal solution for long-duration cargo and exploration missions beyond Mars, where solar flux and chemical propellant limits are critical constraints

III. REACTOR TECHNOLOGIES FOR NEP

NEP systems rely on compact, space-qualified reactors to provide continuous electrical power for electric thrusters during extended deep-space missions. The choice of reactor design critically impacts system performance, mass, reliability, and mission safety. This section reviews the major reactor classes considered or implemented for

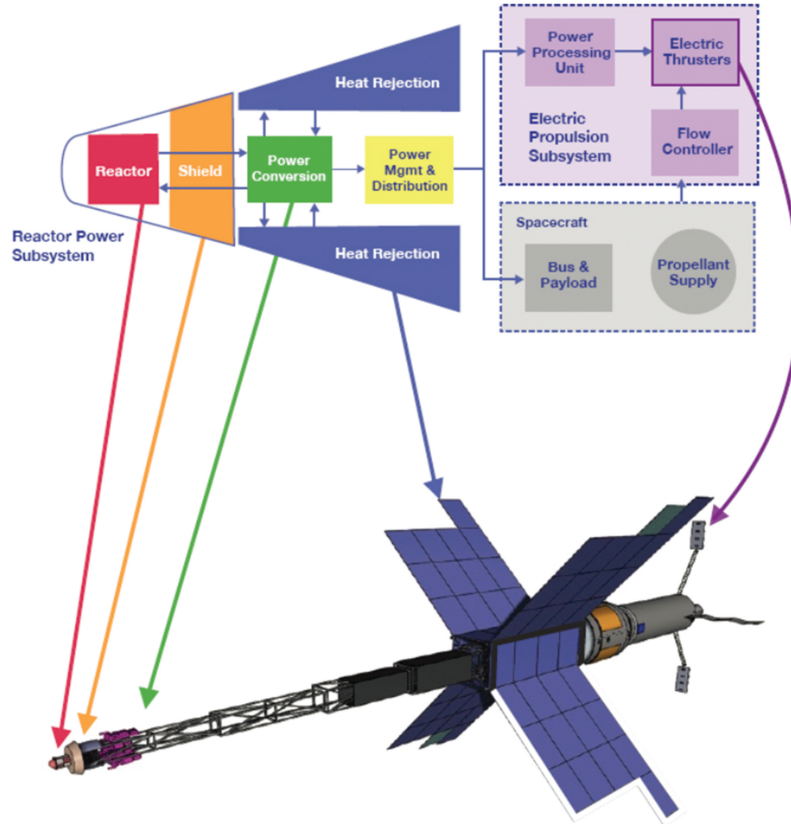


Fig. 2. Schematic architecture of a basic NEP spacecraft illustrating the reactor core, radiator panels, power conversion system, and electric thrusters, along with the heat rejection and power distribution subsystems [1].

TABLE 2
Comparative Analysis of Representative Propulsion Systems [10,11,28–33]

Parameter	Chemical Propulsion	Solar EP	NEP
Isp	~450 s	1500 to 3000 s	3000 to 10 000 s
Thrust	10^5 to 10^6 N	0.1 to 1 N	0.1 to 5 N
Propellant mass	High	Moderate	Low
Payload capacity	High	Moderate	High
Mission duration	Days to months	Months to years	Years
Operational domain	Near Earth	Cislunar/Mars orbit	Deep space/outer planets
Cost per kilogram of payload	High	Medium	Low (long term)

NEP applications, focusing on their operational principles, maturity, and applicability to spaceflight.

Fig. 3 illustrates a conceptual layout of a NEP system, highlighting the functional integration of its core subsystems. The compact nuclear reactor serves as the primary energy source, followed by energy conversion units and electric thrusters arranged to optimize power flow and thermal management. Radiation shielding is positioned to protect the payload or crew modules from harmful emissions, while the radiators are extended outward to enhance heat rejection efficiency. This configuration reflects the essential design considerations for sustained deep-space propulsion.

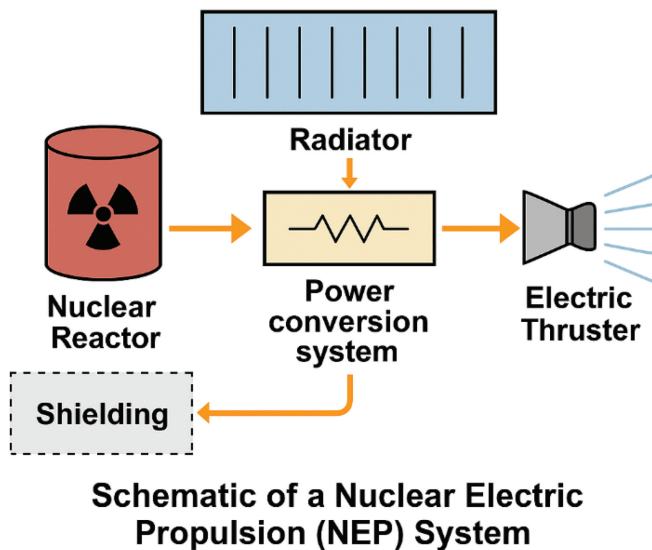


Fig. 3. Schematic representation of a NEP system highlighting the principal subsystems: a compact nuclear reactor for primary power generation, a power conversion unit (e.g., thermoelectric or Brayton cycle), an electric thruster (ion or Hall-effect type), thermal radiators for heat rejection, and radiation shielding for crew and sensitive electronics. The arrows denote thermal energy transfer from the reactor to the power conversion unit, electrical power delivery to the thruster, and dissipation of excess heat through the radiators.

III.A. Solid-Core Reactors

Solid-core reactors are the most established and space-ready class of fission systems developed for propulsion and power applications. These designs rely on fissile fuels, such as ^{235}U or ^{239}Pu , embedded within durable structural matrices that can withstand extreme thermal loads and intense radiation environments. Heat generated by fission is conducted to the power conversion stage, which in compact configurations is typically achieved through thermoelectric modules or Stirling engines [12].

A representative example is NASA's Kilopower reactor, engineered for both surface operations on the Moon and Mars, as well as deep-space missions. Kilopower employs a highly enriched ^{235}U fuel element, with passive sodium heat pipes for thermal transport and Stirling engines to deliver between 1 and 10 kW(electric) of electrical output [12]. The design further incorporates a beryllium oxide neutron reflector and lithium hydride shielding to minimize radiation exposure for onboard electronics and potential crew members [15].

Solid-core reactors offer several advantages, including structural simplicity, demonstrated operational lifetimes, and inherent passive safety features, such as negative temperature coefficients. Their limitations, however, stem from the thermal tolerances of structural materials and the difficulty of rejecting waste heat in the vacuum of space. These factors restrict achievable power densities when compared to more advanced reactor concepts [13].

III.B. Thermionic and Brayton Cycle Reactors

Thermionic reactors generate electricity by exploiting the thermionic effect, in which electrons are emitted from a heated cathode and collected by a cooler anode [21]. Operating at temperatures exceeding 1200 K, these reactors are inherently compact and employ solid-state

configurations with no moving parts, offering high reliability for space applications.

The Soviet TOPAZ program, particularly the TOPAZ-1 and TOPAZ-2 systems, provides well-documented examples. These reactors, fueled with highly enriched uranium (HEU), produced electrical outputs in the tens of kilowatts. TOPAZ-2 underwent extensive ground-based testing and was eventually deployed aboard reconnaissance satellites in the late twentieth century [15,21].

In contrast, reactors employing the closed Brayton cycle convert thermal energy into electricity by circulating a high-temperature working fluid, typically a helium-xenon mixture, through a turbine mechanically coupled to an electrical generator. These systems deliver thermal-to-electric conversion efficiencies in the range of 20% to 30%, surpassing those of thermoelectric modules, and are scalable to the megawatt class [23].

While the SNAP-10A mission relied on thermoelectric conversion, it provided valuable operational experience that informed later Brayton-based designs pursued by NASA and the U.S. Department of Energy (DOE) for advanced NEP platforms [34]. Brayton systems offer notable advantages in efficiency and power scalability for high-demand missions; however, the inclusion of moving turbomachinery introduces added mechanical complexity and potential reliability risks, particularly for multiyear operation in the harsh space environment [23].

III.C. Fission Fragment Reactors and Emerging Concepts

Fission fragment reactors represent an advanced but still exploratory approach within NEP research. These systems aim to achieve very high Isp values ($I_{sp} > 100\,000$ s) by directing the kinetic energy of charged fission fragments outward to produce thrust, bypassing the intermediate step of converting heat into electricity [14]. Concepts under investigation include dusty plasma configurations and direct electrostatic acceleration methods, both of which seek to minimize thermal losses and improve system efficiency.

Despite their theoretical promise, such reactors remain at an early stage of maturity (TRL < 3), with significant technical barriers to overcome. Key challenges include maintaining reactor containment, managing charged particle trajectories, ensuring material durability under extreme radiation conditions, and sustaining operational stability over long mission durations. Continued research and experimental validation are required before fission fragment reactors can progress toward practical application in space propulsion [14].

TABLE 3
Technical Specifications of Key Historical NEP Reactors (Adapted from Refs. [35] and [36])

Reactor System	Type	Fuel Type	Thermal Power	Electric Power	Conversion Method	Mass (approximate)	Status	Notes
SNAP-10A	Solid core	^{235}U , ZrH	~30 kW(thermal)	~500 to 600 W(electric)	Thermoelectric	~290 kg (core)	Launched in 1965, orbited 43 days	First reactor in orbit
TOPAZ-2	Thermionic	^{235}U , UO_2	~115 kW(thermal)	~5 to 10 kW(electric)	In-core thermionic	~320 kg	Ground-tested (TOPAZ-1 flown)	Multicell Thermionic Fuel Element design
Kilopower (KRUSTY) SP-100 (concept)	Solid core	^{235}U (HEU)	4–43 kW(thermal) Up to 2.4 MW (thermal)	1 to 10 kW(electric) ~100 kW(electric)	Stirling engine	Prototype stage ~3000 kg	Ground-tested 2017 to 2018 Cancelled concept	Passive heat pipes DOE/NASA prototype design
Fission fragment reactor	Conceptual	^{235}U uranium nitride (UN) fuel $^{233}\text{U}/^{235}\text{U}$	—	~100 kW(electric) (theoretical)	Thermoelectric/Brayton Direct particle acceleration	—	Research phase	Ultrahigh Isp concept

Table 3 provides the technical specifications for these NEP reactors.

IV. ELECTRIC PROPULSION SYSTEMS

Electric propulsion (EP) systems play a central role in NEP architectures by converting electrical energy into kinetic energy to accelerate the propellant. Unlike chemical propulsion, which delivers high thrust over short durations, electric thrusters operate continuously at high Isp and high fuel efficiency, enabling long-duration deep-space missions. This section highlights the major EP technologies relevant to NEP platforms and compares their performance characteristics and development status.

IV.A. Ion Propulsion

Ion propulsion systems operate by ionizing a neutral propellant gas, typically xenon, and accelerating the ions through electrostatic fields to produce thrust. These systems are known for their extremely high Isp (typically 3000 to 10 000 s), making them ideal for missions requiring fuel efficiency over extended periods.

NASA's NSTAR (NASA Solar Technology Application Readiness) ion engine, used aboard the Deep Space 1 mission, demonstrated the long-duration viability of ion propulsion in interplanetary travel. The more advanced NEXT (NASA Evolutionary Xenon Thruster) engine offers improvements in power handling (up to 7 kW), Isp (~4190 s), and operational lifetime (>50 000 h), making it a strong candidate for future NEP systems.

However, ion engines typically produce very low thrust (in the milli-Newton range) and require high-voltage power conditioning, making thermal control and power scaling key engineering challenges.

IV.B. Hall-Effect Thrusters

HETs utilize a combination of radial magnetic fields and axial electric fields to accelerate ions, producing higher thrust levels than ion engines, though at somewhat lower Isp's (~1500 to 3000 s). The simplicity of their design and robustness have made them attractive for both orbital and interplanetary applications.

The Russian space program has extensively utilized Hall thrusters on satellites and Mars missions. The European Space Agency (ESA) has also adopted HETs, such as the SPT-140, for missions like BepiColombo. Their higher thrust density makes Hall thrusters particularly promising for near-term NEP missions with

moderate power levels (1 to 20 kW). The main drawbacks of HETs include moderate erosion of the channel walls, which limits operational life, and lower efficiency compared to ion thrusters (~50% to 70%).

IV.C. MPD Thrusters

MPD thrusters accelerate plasma using the Lorentz force generated by the interaction of electric currents and magnetic fields. They offer very high thrust densities and are capable of operating at megawatt power levels, making them conceptually well suited for high-thrust NEP applications.

However, MPD systems currently suffer from low overall efficiency (10% to 40%) and require advanced electrode materials to manage intense thermal and electrical loads. While demonstrated in laboratory settings, they remain in early developmental stages and have not yet been flight qualified. If these challenges are resolved, MPD thrusters could serve as propulsion cores for cargo tugs or heavy payload transfers in the outer solar system.

IV.D. VASIMR System

VASIMR is a unique EP system that heats a propellant (typically argon or hydrogen) to plasma state using radiofrequency waves and then accelerates the plasma via magnetic nozzles. Unlike traditional EP systems with fixed parameters, VASIMR allows for real-time adjustment between high-thrust and high-efficiency modes by varying the Isp (2000 to 12 000 s).

Developed by the Ad Astra Rocket Company, VASIMR has undergone ground tests with demonstrated power levels above 100 kW (e.g., VX-200). Its scalability and variable performance profile make it attractive for future NEP missions, such as rapid cargo transfer to Mars. However, challenges include complex Radio Frequency heating systems, large radiator requirements, and the need for high power availability making it an ideal candidate for nuclear-powered platforms rather than solar-electric ones. Table 4 and Fig. 4 provide a performance comparison of the various EP technologies.

Fig. 4 categorizes various EP systems, ranging from resistojets to field-emission electric propulsion (FEEP), while also indicating their throttling capabilities (denoted by $n = 10\%$ to 100%).

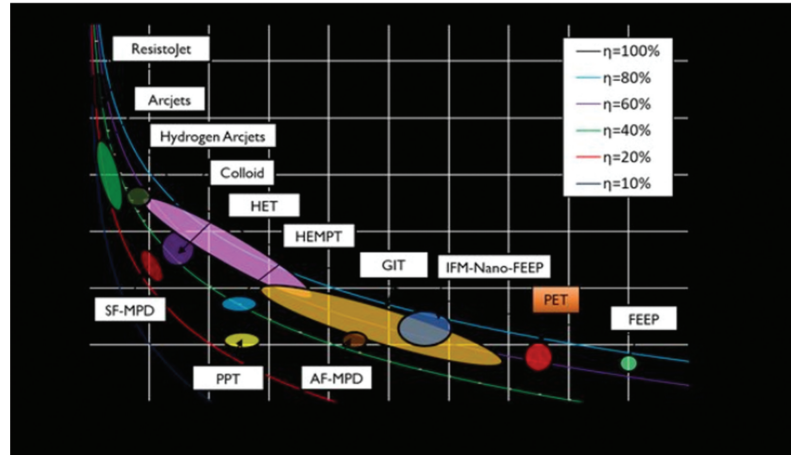
IV.D.1. Key Components of the Figure

Fig. 4 illustrates the classification of major electric propulsion (EP) technologies considered for space

TABLE 4

Performance Comparison of the Electric Thruster Technologies [4–8,15–17,24–27,35–40]

Thruster Type	Isp (s)	Thrust (N)	Efficiency (%)	Power Range (kW)	TRL
Ion (NSTAR/ NEXT)	1900 to 3100 NSTAR; 1320 to 4190 NEXT	0.019 to 0.092 NSTAR; 0.05 to 0.237 NEXT	~60% to 68% at high power	0.5 to 7 kW	High (TRL 8 to 9)
Hall Effect (SPT- 140, PPS-X)	1800 SPT-140 to 2800 to 3000 PPS-X	0.28 N SPT-140; ~0.25 N (5 kW) PPS-X	~56% to 70%	0.9 to 5 kW	High (TRL 7 to 9)
MPD	2000 to 5000 s	~2.5 to 25 N at 100 to 500 kW	40% to 60% (some up to ~61.9%)	100 to 500+ kW	Medium (TRL 3 to 5)
VASIMR (VX-200)	~1660 to 4900 s (varies with power)	Up to ~5.8 N at 200 kW	~72% at 200 kW laboratory test	15 to 200+ kW	Medium (TRL 4 to 6)

Fig. 4. Performance comparison of various EP technologies (Resistojet to FEED) at different throttling levels ($\eta = 10\%$ to 100%) [2].

applications, organized according to their operating principles and performance characteristics. The figure spans low-temperature electrothermal devices to advanced plasma-based and field-emission systems, highlighting the wide spectrum of thrust levels, specific impulse, and throttling capability. The following list briefly describes each propulsion category shown in Fig. 4, providing context for their working principles, typical propellants, and representative application domains.

1. Resistojet (top section):

a. A resist jet is a simple EP system that heats a propellant (often a gas like hydrazine or ammonia) using an electric resistor, improving efficiency over cold-gas thrusters.

b. The throttling levels ($\eta = 10\%$ to 100%) suggest that resistojets can operate at different power settings, adjusting thrust output.

2. Arcjets:

a. Arcjets use an electric arc to superheat propellant (e.g., hydrogen, hydrazine), providing higher Isp values than resistojets.

b. Hydrogen arcjets are a subtype optimized for hydrogen propellant, offering even better performance.

3. Electrostatic and electromagnetic thrusters:

a. Colloid thrusters use charged liquid droplets (colloids) accelerated by electric fields.

b. HETs ionize the propellant (usually xenon) and accelerate the ions using a Hall current.

c. High-efficiency multistage plasma thruster is a variation of ion/plasma thrusters with improved efficiency.

d. Gridded ion thruster uses electrostatic grids to accelerate ions for high-efficiency propulsion.

4. Field Emission and other advanced concepts:

a. Indium FEEP thruster is a miniaturized FEEP variant using indium as propellant.

b. FEEP, which emits ions from liquid metal (e.g., cesium or indium) under high electric fields, is used for ultra-precise thrust.

5. High-power systems:

a. Self-field (SF) MPD thruster, which uses Lorentz force to accelerate plasma, is suitable for high-thrust applications.

b. Applied-field MPD thruster is similar to the SF MPD but with an external magnetic field for better control.

c. Pulsed plasma thruster (PPT) generates short bursts of plasma, often used for attitude control.

d. Pulsed electrothermal thruster is a hybrid concept combining arcjet and pulsed operation.

IV.D.2. Throttling ($N = 10\%$ to 100%)

Description of Throttling Capability in Fig. 4

- Throttling range ($n = 10\%$ – 100%) indicates the ability of each electric propulsion system to vary its thrust output relative to its nominal operating condition.
- Resistojets exhibit a wide throttling range, allowing operation from low-thrust to full-thrust modes.
- Low-power electric propulsion systems, such as field-emission electric propulsion (FEEP) and pulsed plasma thrusters (PPTs), provide fine thrust control but operate at relatively low thrust levels.

- High-power propulsion systems, including arcjets and magnetoplasmadynamic (MPD) thrusters, deliver higher thrust but typically offer more limited throttling flexibility.

- Overall, Fig. 4 highlights the trade-off between thrust adjustability and achievable thrust across different electric propulsion classes.

1. The $n\%$ values likely indicate the throttling capability (ability to vary the thrust levels) of each system.

2. Resistojet shows full throttling (100% down to 10%), while other systems may have more limited throttling ranges.

IV.D.3. Interpretation

1. Fig. 4 likely compares different EP technologies in terms of thrust adjustability and classification.

2. Low-power systems (e.g., FEEP, PPT) may have fine throttling but low thrust, while high-power systems (e.g., MPD, arcjets) offer higher thrust but less throttling flexibility.

V. SYSTEM INTEGRATION AND THERMAL MANAGEMENT

Efficient system integration and robust thermal management are essential for the reliability and longevity of NEP spacecraft. These systems must ensure that electrical power is delivered cleanly to all subsystems, that waste heat from the reactor and power electronics is effectively dissipated in the space environment, and that structural elements support stability during launch and operation.

V.A. Power Conditioning and Distribution

The electrical output from nuclear reactors, often in the form of high-voltage, variable-frequency alternating current, requires precise conditioning before it can be used by propulsion systems and onboard electronics. This is achieved using direct current to direct current (DC to DC) converters, inverters, and regulation units that tailor voltage and current to the needs of the specific subsystems.

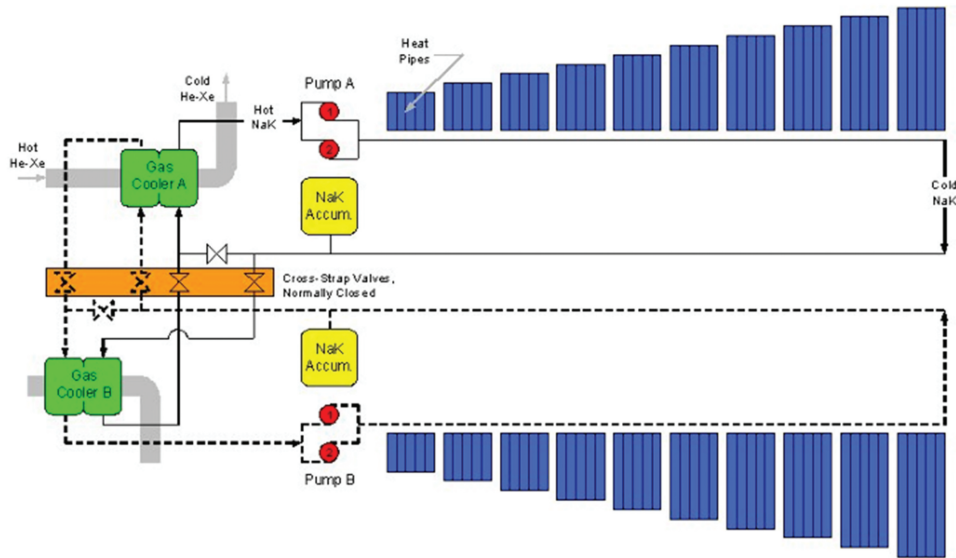


Fig. 5. Schematic representation of the heat rejection subsystem in a NEP spacecraft. The diagram illustrates the integration of a closed-loop heat transport system using NaK coolant, Brayton cycle power conversion, and dual deployable radiator panels with embedded heat pipes for efficient thermal dissipation in space. Structural trusses support radiator symmetry and stability during operation [3].

V.A.1. Key Considerations

To ensure reliable and long-duration operation of nuclear-powered electric propulsion systems, several system-level considerations must be addressed at the spacecraft power architecture level. These considerations govern operational robustness, survivability in harsh space environments, and efficient power delivery to critical subsystems. The following key aspects outline the primary design drivers influencing system reliability and performance.

1. *Redundancy*: Redundant power pathways ensure continued operation in the event of partial system failure.
2. *Shielding*: Radiation-hardened power electronics and electromagnetic shielding prevent performance degradation due to ionizing radiation.
3. *Load management*: Smart power distribution systems dynamically allocate power based on real-time mission demands and system health diagnostics.

V.B. Heat Rejection Systems

Effective thermal management is essential for maintaining the operational stability and longevity of nuclear electric propulsion systems. In the space environment, where convective heat transfer is absent, dedicated

thermal control strategies are required to dissipate reactor and power conversion heat loads. The following approaches represent commonly employed methods for managing heat rejection in space-based nuclear systems.

Nuclear reactors and EP components generate significant thermal loads, especially in the vacuum of space where convection is absent. Therefore, the spacecraft must rely entirely on radiative heat rejection.

V.B.1. Thermal Control Strategies

1. *Radiator panels*: Large-area radiators dissipate heat via thermal radiation. Their size and surface coatings are tailored to maximize emissivity while minimizing absorptivity.
2. *Pumped-loop systems*: Heat is transferred from high-temperature components (e.g., reactor core, Brayton cycle heat exchangers) using fluid loops to radiators. These can be single-phase (liquid coolant) or two-phase (boiling-condensation cycles) systems.
3. *Thermal isolation*: The structural design separates hot and cold regions, preventing heat backflow into sensitive areas like electronics and payload modules.

Fig. 5 illustrates a conceptual layout of the heat rejection subsystem designed for a NEP spacecraft. The system employs a closed-loop thermal management approach,

where a sodium-potassium (NaK) eutectic alloy serves as the coolant. This choice is driven by NaK's excellent thermophysical characteristics and its ability to remain in a liquid state under the extreme temperature variations of space [3].

Thermal energy extracted from the reactor core is converted into electrical power via Brayton cycle turbines. The residual heat is subsequently directed to dual deployable radiator arrays, which are equipped with embedded loop heat pipes to enhance thermal transport efficiency. The radiator panels are optimized for maximum emissivity and reduced structural mass by utilizing high surface area designs and advanced metamaterial coatings. Supporting trusses ensure structural rigidity and proper deployment alignment, enabling stable operation of the radiators in microgravity. This subsystem configuration is vital for sustaining the thermal balance of megawatt-class NEP spacecraft, ensuring safe reactor operation and prolonging mission lifetime in the vacuum of deep space.

V.C. Structural Integration and Vibration Damping

Integrating the NEP system into a spacecraft requires attention to mechanical loads, vibrational harmonics, and thermal expansion.

V.C.1. Key Aspects

The integration of nuclear power systems with electric propulsion architectures introduces additional design constraints related to system scalability, safety, and mission adaptability. Addressing these factors is critical for enabling sustained deep-space missions and autonomous operation. The following points summarize the principal considerations associated with system integration and mission deployment.

1. *Structural trusses*: High-strength, lightweight materials (e.g., carbon composites or titanium alloys) are used to support reactor mass and radiator arrays.
2. *Launch survival*: Components must withstand vibrational and acceleration loads during launch, necessitating shock absorbers or dampers.
3. *Vibration damping*: Electric thrusters, particularly MPD and VASIMR types, may introduce microvibrations. These are mitigated using passive dampers

or active control mounts to maintain spacecraft attitude and pointing accuracy.

VI. MISSION CASE STUDIES

VI.A. Historical Missions

Early demonstrations of nuclear power systems in space were largely experimental, focusing on validating core technologies under orbital conditions. The SNAP-10A mission, launched in 1965, was the first and only U.S. nuclear reactor deployed in orbit [41]. This system utilized a compact fission reactor coupled with thermoelectric energy conversion to supply electrical power for satellite operations. Despite its premature shutdown after 43 days due to a nonnuclear electrical failure, SNAP-10A delivered invaluable data on reactor behavior, heat rejection, and radiation shielding in microgravity [41].

At the same time, the Soviet Union advanced its nuclear space program with the TOPAZ series of reactors, notably TOPAZ-1 and TOPAZ-2 [42]. These thermionic power systems flew on multiple reconnaissance satellites and demonstrated reliable electricity generation in space environments. The TOPAZ-2 unit was later transferred to the United States under the STS-91 mission for evaluation, symbolizing a rare collaborative effort between Cold War adversaries [42].

These pioneering missions laid critical groundwork for understanding reactor operation, thermal regulation, and shielding strategies in space, although they did not include fully integrated NEP propulsion systems.

VI.B. Planned and Future Missions

Contemporary space programs are moving toward operationalizing NEP systems for long-duration exploration. NASA's DRACO, developed in collaboration with the U.S. Department of Defense's (DoD's) Defense Advanced Research Projects Agency (DARPA), integrates nuclear propulsion into a highly maneuverable spacecraft designed for agile logistics in cislunar space [30]. While DRACO primarily focuses on NTP, studies by Hofer et al. [31] suggest its architecture could be adapted for NEP applications in future deep-space missions requiring sustained low-thrust operation.

In Europe, the ESA has initiated feasibility studies on nuclear-powered space tugs aimed at interplanetary cargo transport [33]. These studies, conducted under the Advanced Research in Telecommunication Systems (ARTES) and Future Launchers Preparatory Programme

TABLE 5
Representative Mission Profiles Enabled by NEP [10,11,20,32,45–48]

Mission Type	Example/Concept	Payload Mass (kg)	Power Level [kW(electric)]	Thruster Type	Estimated Isp (s)	Thrust (N)	Mission Duration
Demonstration/technical demonstration	Kilopower + Hall/ion concept	~500	1 to 10	Hall effect/ion	2000 to 4190	0.1 to 0.3	~1 year
Cislunar logistics tug	DRACO-derived NEP	2000 to 4000	50 to 100	Ion/MPD	2500 to 5000	0.5 to 2	6 months to 1 year
Mars cargo transfer	NEP tug with VASIMR/MPD	10 000+	200 to 300	VASIMR/MPD	5000 to 7000	5 to 20	3 to 6 months (Earth to Mars)
Asteroid prospector	Long-duration robotic platform	1000 to 2000	40 to 80	Ion/Hall effect	3000 to 5000	0.3 to 1	1.5 to 3 years
Jovian system orbiter	Jupiter Icy Moons Orbiter (Prometheus concept)	3000 to 6000	~200	Ion + Brayton-powered	7000 (Prometheus)	1 to 5	~5 to 7 years
Mars sample return tug	Hall/VASIMR return vehicle	5000 to 7000	150 to 200	Hall effect/VASIMR	3000 to 6000	2 to 5	~18-month round trip

(FLPP) programs, explore integrating EP technologies with compact nuclear power sources to enhance mission endurance [43].

Similarly, the China National Space Administration (CNSA) is pursuing long-term ambitions for Mars-bound robotic and crewed missions using compact fission reactors as primary energy sources for EP [44]. These developments are built on CNSA's Chang'e lunar program and Tianwen Mars missions, which have established operational expertise in long-duration deep-space systems [44].

Beyond DRACO, NASA is also exploring NEP integration within the Artemis program. Autonomous cargo platforms utilizing NEP could serve as logistical bridges between Earth orbit and lunar or Martian staging points, reducing propellant consumption and increasing mission frequency [31].

VI.C. Proposed Interplanetary Use Cases

The application of NEP systems extends to various interplanetary mission architectures. A prominent example is a Mars cargo tug capable of transporting large infrastructure components and crew habitats with reduced propellant mass compared to chemical propulsion systems [1]. NEP platforms are also envisioned for asteroid resource extraction missions.

By supporting extended stays near near-Earth objects, NEP-enabled spacecraft could facilitate robotic and semi-autonomous material harvesting and return operations [37].

In the outer solar system, NEP systems present a viable means of sustaining operations within the challenging environments of the Jovian system. Missions targeting Europa or Ganymede would benefit from NEP's ability to provide continuous power and propulsion under high-radiation conditions and low solar flux [32]. Such systems could enable orbiters, cryobots, or subsurface probes to achieve unprecedented mission durations and scientific return. Table 5 provides profiles of representative missions enabled by NEP, and Table 6 gives the proposed NEP mission taxonomy by power class. Fig. 6 illustrates the timeline of key milestones in the development and deployment of NEP.

VII. GLOBAL PROGRAMS

The development and deployment of NEP systems are being actively pursued by multiple spacefaring nations, each leveraging unique technological pathways and strategic priorities. This section outlines the key programs and international initiatives currently shaping the global NEP landscape.

TABLE 6
Proposed NEP Mission Taxonomy by Power Class*

Power Class	Typical Applications	Example Systems
<10 kW(electric) 10 to 100 kW(electric) >100 kW(electric) (megawatt class)	Lunar surface systems, small satellites Cislunar cargo transfer, Mars robotics Crewed Mars transit, Jovian exploration	SNAP-10A, Kilopower DRACO (planned), TOPAZ Conceptual TEM, VASIMR

*Source: Authors' classification based on mission profiles and reactor scaling studies [12,36,49].

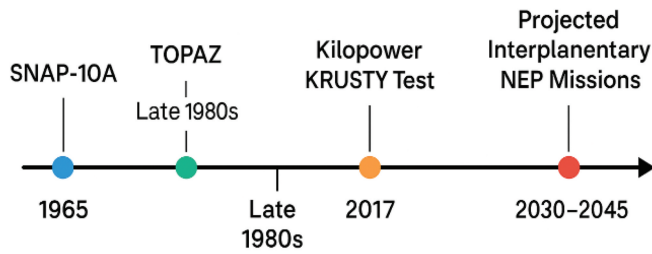


Fig. 6. Timeline of key milestones in the development and deployment of NEP systems, including historical reactors such as SNAP-10A and TOPAZ, current programs like NASA's Kilopower and DRACO, and projected interplanetary mission implementations extending into the 2030 to 2045 timeframe.

VII.A. NASA Initiatives: Kilopower and DRACO

*NASA has been at the forefront of NEP research through initiatives like the Kilopower project, which focuses on developing compact, scalable solid-core fission reactors capable of supporting lunar, Martian, and deep-space missions [10,23]. Kilopower reactors utilize ^{235}U fuel and passive heat pipe systems coupled with Stirling engines to deliver electrical power in the 1- to 10-kW range [28].

Ground tests, such as the Kilopower Reactor Using Stirling Technology (KRUSTY), have validated operational feasibility and fuel efficiency under simulated space conditions [50]. Parallel to Kilopower, NASA, in collaboration with DARPA, is advancing the DRACO project. Although DRACO primarily focuses on NTP, its hybrid architecture and potential adaptability are relevant for future NEP configurations [31]. DRACO aims to enable agile maneuvering and logistics across cislunar and interplanetary domains [30].

VII.B. Roscosmos and Russia's Nuclear Tug

Russia's Roscosmos agency is developing the Transport and Energy Module (TEM), also known as Zeus, is a nuclear-powered space tug designed for long-duration deep-space missions [33]. The TEM is expected

to incorporate a 500 kW-class nuclear power system paired with EP technologies, likely HETs or MPD thrusters [42]. Early reports indicate plans for a flight demonstration in the early 2030s, positioning Russia as a leader in large-scale NEP deployment [33].

VII.C. ESA Programs

The ESA has progressively expanded its activities in advanced propulsion, with NEP included as part of broader technology development initiatives. Notable examples are the ARTES program and FLPP, which support the maturation of enabling technologies [33,43]. Within these frameworks, ESA has worked with member states on EP systems such as HETs and gridded ion engines, often designed for integration into chemical-electric hybrid platforms [43].

Although ESA does not yet maintain a nuclear propulsion program comparable in scale to those pursued by NASA or Roscosmos, feasibility studies and parallel efforts to establish nuclear safety guidelines indicate a clear interest in contributing to potential international collaborations on future NEP missions [13].

VII.D. China's Advancing Capabilities: CNSA Programs

The CNSA is making rapid progress in nuclear space technologies. Plans reportedly include nuclear-powered lunar surface systems and fission-based propulsion for Mars cargo missions targeted for the mid-2030s [44]. These initiatives are supported by CNSA's broader robotic space programs, including the Chang'e lunar series and the Tianwen-1 Mars orbiter and rover [44]. Although public data remain limited, investments in high-power space reactors and long-duration EP indicate strong strategic ambitions in NEP development [44].

VII.E. International Treaties and Regulatory Landscape

The deployment of nuclear power systems in space is governed by international treaties, national regulations, and

voluntary guidelines. The Outer Space Treaty (1967) prohibits the placement of nuclear weapons, but permits peaceful nuclear technology applications [51]. The United Nations Principles Relevant to the Use of Nuclear Power Sources in Outer Space (1992) provide safety standards and environmental protection measures [52]. Additionally, national-level oversight, such as DOE and U.S. Federal

Aviation Administration licensing, plays a critical role in launch approvals for nuclear systems [49].

As NEP missions transition from conceptual design to operational deployment, addressing legal, safety, and geopolitical concerns will be essential. Updated protocols covering launch authorization, orbital debris management, reactor passivation, and planetary protection are

TABLE 7
International Programs and Initiatives in NEP [19,28–33,44,50,55,56]

Country/Program	Leading Organizations	Description and Status	Timeline
USA Kilopower + VASIMR/ JETSON	NASA, SpaceNukes, Ad Astra Rocket Company	NASA-developed Kilopower small fission reactor [1- to 10-kW(electric) tested via KRUSTY], paired with Ad Astra's VASIMR thruster. Joint partnership aims to demonstrate 100 kW+ NEP in flight and scale toward multi-megawatt NEP systems by the end of the decade	Flight demonstration by ~2030; commercialization in 2030s
ESA Europe RocketRoll Consortium	ESA, Tractebel (Engie), CEA, ArianeGroup, OHB Czechspace, CTU Prague, IRS Stuttgart, Airbus	Funded under ESA's FLPP; completed 2023–2024 feasibility study for NEP tug design; submitted roadmap for a demonstrator mission by 2035	Roadmap finalized >2024; target demonstrator 2035)
Russia TEM	Roscosmos, Rosatom, Keldysh Center, NIKIET	Project TEM aims for a 3.8-MW reactor delivering ~1 MW(electric) power for propulsion; designed as a Zeus-class cargo tug with Isp ~7000 s)	Orbital reactor tests aimed by ~2030 (based on program timelines)
United Kingdom Pulsar Fusion (Sunbird Fusion Drive)	Pulsar Fusion, supported by United Kingdom Space Agency	Fusion-based plasma propulsion; development ongoing with ground tests targeted in the mid-2020s)	Ground testing in 2025 to 2026; in-space demonstration as early as 2027
Historical/informational: Prometheus and Jupiter Icy Moons Orbiter	NASA, DOE, Russian agencies (NIKIET, Keldysh Center)	Jupiter Icy Moons Orbiter/ Prometheus aimed for a ~200- kW(electric) NEP spacecraft to explore Jupiter, but terminated in 2005. TOPAZ and earlier Soviet thermionic designs contributed to modern NEP knowledge base.	2003 to 2005 baseline projects; now historical context
Global/study: JETSON DoD program, U.S. Air Force Research Laboratory, NASA	U.S. Air Force Research Laboratory, NASA	JETSON aims for a 10-kW(electric)-class fission power system for in-space demonstration, combined with NEXT-C or Hall thrusters for deep- space science missions	Technology development underway; integrated flight demonstration envisioned within late 2020s

urgently needed for interplanetary missions using fission-based power systems [53,54].

VII.F. Experimental Validation and Flight Demonstrations

Although NEP systems have largely been developed through theoretical studies and subsystem testing, recent and planned experimental programs demonstrate their practical viability. These efforts bridge the gap between laboratory-scale prototypes and mission-ready spacecraft.

VII.G. Ground-Based Reactor Tests

NASA's Kilopower initiative achieved a major milestone with the KRUSTY test in 2018 at Nevada's national nuclear site. It sustained a 1-kW(electric) output using a solid-core ^{235}U reactor paired with passive sodium heat pipes and Stirling conversion, validating scalable designs for 1- to 10-kW(electric) NEP systems.

VII.H. Electric Thruster Validation

Ad Astra's VASIMR VX-200 thruster attained power levels of up to 200 kW in laboratory testing, enabling variable specific impulse in the range of approximately 1500 to 5000 s, significantly exceeding the specific impulse achievable by nuclear thermal propulsion systems such as NERVA. Similarly, NASA's NEXT ion thruster logged over 48,000 operational hours in endurance testing, reinforcing its suitability for long-duration missions.

VII.H.1. Planned Flight Demonstrations

Several planned and proposed flight demonstration programs are expected to play a critical role in advancing the technology readiness of nuclear-based space propulsion systems. These initiatives aim to validate reactor operation, power conversion, and propulsion integration in relevant space environments, thereby bridging the gap between ground testing and sustained operational deployment. The following programs represent key near- to mid-term efforts toward demonstrating nuclear thermal and nuclear electric propulsion capabilities.

1. *NASA/DARPA DRACO*: This joint program aims to demonstrate nuclear thermal rocket propulsion in cislunar space as early as 2027. DRACO's architecture, while primarily NTP, is engineered for potential NEP hybridization in subsequent missions. The program could mark

a transformational step in technology readiness for deep-space propulsion. Note: There have been recent delays due to reactor ground-testing requirements.

2. *Roscosmos TEM*: TEM is a nuclear-electric space tug concept based on a gas-cooled high-temperature reactor [~ 3.8 MW(thermal), ~ 1 MW(electric)] capable of powering high-thrust EP. Projected for initial deployment around 2030, it represents a clear leap in NEP readiness for orbital servicing and interplanetary logistics.

VII.H.2. Implications for NEP Deployment

These experimental programs are key to advancing NEP technology from theoretical models to functional systems:

1. *TRL advancement*: Ground demonstrations (KRUSTY, VASIMR, NEXT) raise components to TRL 5 to 6.

2. *Operational validation*: DRACO and TEM offer pathways to TRL 8 to 9, proving technologies in relevant space environments.

3. *Stakeholder confidence*: Demonstrations help de-risk missions scientifically, technically, and politically, aligning them with exploration goals and ethical/public acceptance.

Table 7 provides a consolidated comparison of leading national efforts in the development of NEP systems. These programs vary in terms of propulsion methodologies, reactor designs, and expected mission timelines. The United States and Russia are progressing toward the deployment of advanced fission-based systems, with some nearing the demonstration phase. Meanwhile, the ESA is coordinating multinational efforts under the RocketRoll initiative to achieve a NEP test bed by 2035. The United Kingdom is investigating the feasibility of fusion-based space propulsion. Legacy programs, such as NASA's Prometheus and Russia's TOPAZ, have laid the groundwork for many of today's ongoing technological advancements.

VIII. CHALLENGES AND LIMITATIONS

NEP systems hold significant potential for enabling high-efficiency, long-duration missions beyond Earth orbit. However, their widespread implementation is constrained by a combination of engineering, biological, regulatory, and risk-related challenges. Addressing these

barriers is critical for ensuring the safe, reliable, and globally acceptable use of nuclear technology in space exploration.

VIII.A. Reactor Miniaturization and Space Qualification

A major technical hurdle in NEP development is designing reactors that are compact and lightweight while maintaining robust power output and operational safety. Space-qualified reactors must withstand extreme thermal gradients, mechanical stresses, and vacuum conditions over multiyear missions [17]. Achieving high power densities in a compact architecture requires innovations in heat management, structural materials, and autonomous control systems.

NASA's Kilopower program demonstrated the feasibility of small-scale reactors [~ 1 - to 10-kW(electric)] for lunar and Martian applications using ^{235}U fuel, passive heat pipe cooling, and Stirling engines for energy conversion [12]. Scaling such designs to 100 kW(electric) or beyond for crewed interplanetary missions introduces additional complexities in thermal management, power conversion efficiency, and system integration [47]. Modular reactor arrays have been proposed as a strategy for incremental scaling and redundancy.

VIII.A.1. Radiation Shielding Materials

Advanced shielding solutions are under development to reduce mass while maintaining protection against ionizing radiation. Materials such as boron carbide, lithium hydride, tungsten composites, and graphene-based multilayers exhibit high neutron attenuation and mechanical resilience, making them suitable candidates for NEP shielding systems [57]. These materials aim to meet International Commission on Radiological Protection dose limits for human-rated missions.

VIII.A.2. Reactor Startup in Microgravity

Starting a nuclear reactor in microgravity environments poses challenges related to fluid management, neutron flux stabilization, and thermal regulation. Kilopower's passive sodium heat pipe design demonstrated reliable startup characteristics under ground-simulated microgravity [12]. Meanwhile, ESA studies in 2024 explored active fluid management techniques to enhance startup reliability in dynamic orbital conditions [58].

VIII.A.3. Advanced Heat Rejection Systems

Efficient heat rejection is critical for high-power NEP systems operating in space's thermal vacuum. Emerging technologies, such as deployable metamaterial radiators and loop heat pipes, have been proposed to improve thermal efficiency and reduce system mass for megawatt-class applications [59]. These solutions are designed to handle the high heat fluxes associated with advanced reactor designs.

VIII.B. Radiation Shielding and Biological Risks

Operating nuclear reactors in space inevitably produces ionizing radiation, posing hazards to both electronic systems and biological payloads, including astronauts. Shielding strategies often rely on materials such as boron, lithium hydride, tungsten, and water to optimize mass efficiency and protective effectiveness [57].

In human-rated systems, reactors are typically mounted on extended booms to distance them from crewed sections, reducing radiation exposure. However, this configuration increases structural complexity and may limit spacecraft maneuverability. Secondary radiation from shielding interactions and neutron scatter can further impact sensitive instruments and communication systems [60].

VIII.C. Political and Regulatory Barriers

The deployment of nuclear systems in space is governed by national and international legal frameworks. In the United States, space nuclear launches require environmental and safety assessments from the DOE and presidential authorization under National Security Presidential Memorandum-20 [33]. Globally, the Outer Space Treaty (1967) and the United Nations Principles Relevant to the Use of Nuclear Power Sources in Outer Space (1992) provide guidelines for safe and peaceful utilization [61].

Public perception and geopolitical sensitivities about nuclear technology, particularly regarding risks of orbital contamination and nuclear proliferation, often delay international collaborations. The SNAP-9A failure in 1964, which resulted in ^{238}Pu dispersal in the atmosphere, serves as a historical reminder of these concerns [34].

VIII.D. Risk of Failure and Debris Contamination

Mission failures during launch or early orbital phases represent a significant concern for NEP systems. A catastrophic failure could release radioactive materials

into Earth's atmosphere or create long-lived orbital debris contaminated with nuclear isotopes [43]. Modern NEP designs incorporate multiple fail-safe features, including robust fuel encapsulation, delayed reactor startup until stable orbit is achieved, and reactor passivation protocols at end of life [44].

Nevertheless, residual risks remain, especially regarding uncontrolled deorbiting and debris management. Effective mitigation strategies for nuclear-contaminated debris will be essential for maintaining the long-term sustainability of Earth's orbital environment.

VIII.D.1. Nuclear Security in Launch Failures

The possibility of a launch failure introduces not only technical and environmental risks, but also critical nuclear security challenges. An accident during ascent or reentry could result in the release of radioactive material into populated regions, with potentially severe health, ecological, and geopolitical consequences.

To reduce such risks, contemporary NEP system designs emphasize multilayered containment, accident-tolerant fuel forms, and robust encapsulation to prevent dispersal under extreme conditions. In addition, most reactor concepts are configured for delayed startup, ensuring that fission does not commence until a stable orbit is achieved. These safeguards, aligned with international nuclear safety and security frameworks, are essential for reinforcing public trust and ensuring that space nuclear power is deployed responsibly and sustainably.

VIII.E. Critical Research Gaps: Barriers to Megawatt-Class NEP Deployment

Although NEP has been the subject of theoretical and experimental investigation for over 6 decades, no megawatt-class system has yet transitioned from concept to operational deployment. This persistent gap between extensive research and practical realization arises from several interdependent challenges.

VIII.E.1. Thermal Rejection Constraints

Scaling NEP systems from the kilowatt-to-megawatt range introduces exponential increases in heat rejection demands. For instance, a 1-MW-class reactor operating at typical radiator temperatures (~ 600 K) would require deployable radiator arrays exceeding 1000 m^2 in surface area to maintain thermal equilibrium [49]. The development of advanced high-temperature radiator materials and

metamaterial-based designs is essential to reduce system mass and stowage volume.

VIII.E.2. Regulatory and Political Barriers

The deployment of nuclear power in space remains tightly regulated under international treaties such as the Outer Space Treaty (1967) and the United Nations Principles Relevant to the Use of Nuclear Power Sources in Outer Space (1992). Public concern over orbital contamination and geopolitical sensitivities about nuclear proliferation have historically limited high-power reactor launches. Moreover, no globally accepted safety protocols currently exist for megawatt-class systems [61].

VIII.E.3. Launch Vehicle Limitations

Until recently, the lack of heavy-lift vehicles capable of delivering large, shielded NEP systems to orbit presented a significant bottleneck. Emerging platforms, such as SpaceX's Starship and NASA's Space Launch System, offer the payload capacities necessary for deploying megawatt-class reactors, potentially overcoming this long-standing constraint.

VIII.E.4. Low TRLs

Critical NEP subsystems, including long-life, high-thrust MPD thrusters and advanced Brayton cycle power converters, remain below TRL 5 for megawatt-class applications. These technologies require extensive ground testing and validation under relevant spaceflight conditions before they can support operational missions [53].

VIII.E.5. Path Forward

Bridging these gaps necessitates a coordinated effort involving international regulatory harmonization, strategic private-sector investment, and targeted technology maturation programs. Focused research on modular reactor architectures, autonomous fault management, and debris-safe reactor passivation will be essential to enable the safe and effective deployment of megawatt-class NEP systems for crewed and robotic interplanetary missions.

VIII.F. Public Perception of Nuclear Risks

Public attitudes toward space nuclear systems are heavily influenced by historical incidents, such as the SNAP-9A failure in 1964, which resulted in ^{238}Pu

dispersal into Earth's atmosphere. Concerns over potential launch failures, reactor accidents, and contamination of celestial bodies continue to shape public discourse. Building public trust requires transparent risk assessments, open communication strategies, and international dialogue to address societal apprehensions about nuclear technologies in space [61].

VIII.G. Orbital Debris and Planetary Protection

The end-of-life management of NEP systems raises additional ethical questions. Uncontrolled deorbiting of reactors or failures during orbital operations could create radioactive debris, posing long-term hazards to both active satellites and future missions. To mitigate these risks, NEP designs now incorporate passive safety mechanisms such as delayed reactor startup until after orbit insertion and end-of-mission passivation protocols. Furthermore, planetary protection guidelines under the Outer Space Treaty mandate preventing biological contamination of extraterrestrial environments, necessitating strict safeguards when operating nuclear systems near potentially habitable worlds [43].

VIII.H. Gaps in International Governance

Current international frameworks, including the Outer Space Treaty (1967) and the United Nations Principles Relevant to the Use of Nuclear Power Sources in Outer Space (1992), provide general guidance but lack detailed enforcement mechanisms for megawatt-class NEP missions. There is an urgent need for updated agreements that address launch authorization, liability in the event of nuclear incidents, and standardized protocols for reactor disposal in orbit. Expanding the role of the United Nations Committee on the Peaceful Uses of Outer Space or establishing a new global oversight body could

facilitate equitable and safe deployment of space nuclear technologies [44].

VIII.I. Ethical Imperatives for Human Spaceflight

When applied to crewed missions, ethical considerations extend beyond technical safety to include protecting astronauts from radiation exposure, minimizing ecological disruption of extraterrestrial environments, and ensuring that nuclear propulsion development supports equitable access to deep-space exploration capabilities. Collaboration between nations and private entities is essential to prevent the monopolization of nuclear propulsion technologies and to promote peaceful use of space.

VIII.J. Public Engagement Strategies for NEP Acceptance

The successful adoption of NEP systems requires more than technological maturity and regulatory frameworks; it also depends heavily on public trust and societal acceptance. Historical experiences from terrestrial nuclear energy programs emphasize the critical role of proactive public engagement in addressing concerns about safety, environmental risks, and ethical implications.

To foster confidence in NEP missions, space agencies and private operators should adopt transparent communication strategies. These may include publishing independent safety reviews, organizing public consultations, and involving diverse stakeholders in decision-making processes. Such approaches can help counter misinformation and build an informed public consensus.

Educational outreach efforts targeting schools, universities, and media platforms are equally vital for improving public understanding of the scientific

TABLE 8
Comparative Overview of Propulsion Options for Mars Missions*

Parameter	NEP	NTP	Solar Sails
Isp	3000 to 10 000 s	~900 s	>10 000 s
Transit time (Earth–Mars)	6 to 12 months	~6 months	3 to 5 years
Payload capacity	High (10 to 40 tons)	Moderate (10 to 20 tons)	Low (<1 ton)
Mission cost	\$2 to \$4 billion	\$1 to %2 billion	<\$500 million
Technical complexity	High	Moderate	Low
Risk profile	Moderate (thermal, rad)	Moderate (startup)	Low (sail degradation)
Technology readiness	TRL 5 to 6 (subsystems)	TRL 6 to 7	TRL 7 to 8

*Source: Authors' synthesis based on NASA DRA 5.0, ESA Mars exploration studies, and DOE technical reports [10,47,54].

principles and robust safeguards underpinning space nuclear technologies. By cultivating a culture of openness and accountability, these initiatives can mitigate opposition and pave the way for broad societal support of NEP-enabled deep-space exploration.

VIII.K. Radiator Mass Scaling for Megawatt-Class NEP Systems

The mass of radiator systems in nuclear electric propulsion (NEP) architectures increases nonlinearly with reactor power, as governed by the Stefan-Boltzmann law, $Q = \sigma \epsilon A T^4$, where Q is the heat to be rejected, σ is the Stefan-Boltzmann constant, ϵ is emissivity, A is the radiator surface area, and T is the radiator temperature. For a representative radiator temperature of 600 K and an emissivity of 0.85, a 1-MW thermal reactor requires approximately 1,950 m² of radiator area. Assuming a conservative structural mass of 5 kg m⁻², the total radiator mass exceeds 9.7 metric tons. Operating radiators at higher temperatures, such as 1000 K, could reduce the required mass to approximately 3.5 metric tons, highlighting the importance of high-temperature radiator materials for megawatt-class NEP systems [59].

VIII.L. Comparative Analysis of Deep-Space Propulsion Systems

Selecting an optimal propulsion system is critical to mission success in deep-space exploration, requiring careful trade-offs among performance, cost, and operational risk. Nuclear electric propulsion (NEP), nuclear thermal propulsion (NTP), and solar sail systems each offer distinct advantages and limitations, making them suitable for different mission profiles. A broad body of experimental, analytical, and mission-level research has examined electric propulsion technologies including ion, Hall-effect, magnetoplasmadynamic (MPD), pulsed plasma, electrospray, and microthruster systems with emphasis on lifetime characterization, material erosion, system integration, and mission capability for small spacecraft and deep-space applications [63–80].

VIII.L.1. Cost Considerations

NEP systems involve substantial initial investments, estimated at \$2 to \$4 billion per mission, due to reactor development, advanced power conversion technologies, and stringent safety requirements. However, NEP's high efficiency can reduce propellant mass and launch costs

for long-duration logistics. NTP, by comparison, is projected at \$1 to \$2 billion per mission because of its simpler reactor design and higher thrust levels, though its reliance on large propellant loads increases operational complexity. Solar sails represent the most economical option (<\$500 million), leveraging passive propulsion mechanisms, but their limited thrust severely restricts their applicability for high-payload or time-sensitive missions.

VIII.L.2. Risk Assessment

NEP faces challenges in thermal management and radiation protection, particularly for crewed missions. Long reactor-to-crew separation booms and shielding technologies are required to mitigate these risks. NTP systems encounter potential hazards during reactor startup and shutdown under dynamic flight conditions, despite extensive heritage from ground-testing programs like NERVA. Solar sails carry minimal technical risks, but are susceptible to micrometeoroid damage and degradation from prolonged radiation exposure during multiyear missions.

VIII.L.3. Mission Performance

NEP offers superior Isp (3000 to 10 000 s), enabling high efficiency for sustained interplanetary cargo transport and modular assembly missions. NTP achieves faster transits (~6 months to Mars) due to its higher thrust-to-weight ratio, but at the expense of greater propellant mass. Solar sails excel in endurance and require no onboard propellant, but their extremely low thrust results in transit times of 3 to 5 years, making them unsuitable for human exploration or time-critical logistics. A comparative overview of the propulsion options for Mars missions is shown in Table 8.

The estimated costs for NEP-enabled Mars missions (\$2 to \$4 billion) are broadly aligned with, and in some cases slightly lower than, recent flagship space programs. For comparison, NASA's Artemis lunar missions are projected at approximately \$4.1 billion per launch, while the James Webb Space Telescope incurred a total development and deployment cost of nearly \$10 billion.

Although the upfront investment for NEP is substantial, primarily due to reactor research, development, and stringent regulatory compliance, the technology promises significant operational advantages. These include reduced propellant mass requirements, greater payload capacities, and the ability to support higher mission cadence through reusability and long-duration performance.

Conversely, chemical propulsion systems, despite their technological maturity, demand large fuel reserves and multiple orbital assembly operations to support deep-space logistics. This not only increases mission complexity, but also elevates cumulative costs, with estimates ranging from \$5 to \$7 billion per Mars mission.

Solar sail technologies offer the lowest initial cost (<\$500 million), yet their extremely low thrust levels result in extended transit durations of 3 to 5 years, rendering them unsuitable for time-sensitive cargo or human exploration missions.

A comparative cost-benefit analysis thus positions NEP as a strategically balanced solution. While more complex than solar sails and initially costlier than NTP systems, NEP provides the scalability and efficiency needed for sustained interplanetary supply chains and long-duration outer planet missions where chemical and solar propulsion approaches face inherent performance constraints

VIII.L.4. Strategic Implications

NEP systems are particularly well suited for heavy cargo and modular interplanetary infrastructure development, balancing high efficiency with acceptable transit times. NTP remains advantageous for crewed missions

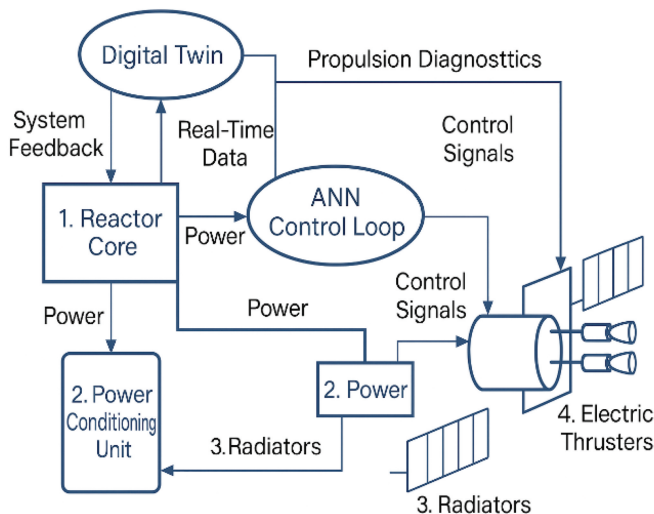


Fig. 7. Conceptual schematic of an AI-integrated NEP spacecraft. The architecture features a closed-loop control system enabled by a digital twin and an ANN that continuously monitor and adjust the spacecraft's reactor output, power conditioning modules, heat rejection system, and electric thruster performance. This intelligent system enables adaptive fault detection, predictive maintenance, and autonomous mission optimization in real time.

requiring rapid transit, but is limited by propellant constraints. Solar sails, with minimal operational costs and infinite endurance, are ideal for exploratory robotic probes targeting distant or long-duration missions where time is not critical. Hybrid propulsion concepts combining NEP with chemical or solar-electric systems may offer synergistic benefits, optimizing payload delivery, mission flexibility, and cost efficiency for future exploration architectures.

IX. FUTURE OUTLOOK AND INTEGRATION WITH AI

As NEP systems move toward operational deployment for deep-space missions, their complexity and longevity demand a paradigm shift in system autonomy, adaptability, and reliability. The integration of AI and artificial neural networks (ANNs) represents a transformative enabler for next-generation NEP platforms. These technologies promise enhanced operational resilience, real-time system optimization, and intelligent fault mitigation, crucial for missions spanning years or decades without direct human intervention.

IX.A. AI/ANN in Propulsion Control

Modern NEP systems involve tightly coupled subsystems, thermal loops, power conversion units, thrusters, and structural components that must remain synchronized under dynamic conditions. AI and ANN-based control architectures can learn complex, nonlinear relationships between propulsion inputs and system responses, enabling the following:

1. Fault detection and anomaly resolution in real time through pattern recognition and predictive modeling.
2. Autonomous thrust vectoring for trajectory correction based on in situ sensor data, mission priorities, and environmental disturbances.
3. Adaptive mission planning, where AI algorithms dynamically adjust power allocation, thrust levels, and thermal loads based on spacecraft health, mission phase, or evolving objectives.

These capabilities are critical for high-stakes missions such as crewed Mars transits or asteroid retrieval, where human-in-the-loop control is impractical due to time delays or communication loss.

IX.B. Digital Twins and Predictive Maintenance

The concept of a digital twin, a real-time, high-fidelity virtual model of the NEP spacecraft, opens new frontiers in diagnostics, control, and maintenance. By continuously ingesting data from onboard sensors, the digital twin mirrors the physical system's behavior, enabling the following:

1. Predictive maintenance, where AI forecasts potential component failures (e.g., in radiators, bearings, or converters) and suggests preemptive actions.
 2. System-level optimization, such as adjusting reactor power or thermal radiator orientation to reduce wear or enhance efficiency.
 3. Root-cause analysis and reconfiguration in the event of subsystem degradation, reducing mission risk and extending operational life.
- Digital twin frameworks, coupled with machine learning algorithms, not only enhance system survivability, but also accelerate design iteration and simulation validation during ground testing.

IX.C. Scalable NEP Architectures and Modular Assembly

The next generation of NEP platforms is envisioned to be modular, upgradable, and scalable, enabling incremental enhancement through orbital assembly or robotic servicing. AI will be critical in managing the dynamic reconfiguration of such systems, especially as mission profiles expand from single-spacecraft operation to distributed NEP-powered fleets or modular Mars cargo trains. Modularity also facilitates mass-efficient launch strategies, where smaller reactor and propulsion modules are assembled and coordinated in orbit, with AI handling system integration, self-calibration, and multinode communication.

Fig. 7 illustrates a conceptual design of a next-generation NEP spacecraft enhanced with AI-driven control systems for deep-space missions. The layout integrates the following key subsystems:

1. *Reactor core*: The spacecraft's nuclear reactor generates thermal energy through fission. This is the primary power source for the entire system.
2. *Power conversion and conditioning*: The reactor heat is converted to electrical energy using systems such

as Brayton cycles, thermoelectric, or thermionics. DC-DC converters and power regulators condition the electrical output for spacecraft systems.

3. *Digital twin*: A real-time virtual replica of the spacecraft operates alongside the physical system. It simulates subsystem behavior, predicts failures, and tests control strategies without affecting the actual system.

4. *ANN*: The ANN processes inputs from sensors, comparing expected versus actual behavior. It enables autonomous decision-making for:

- a. Reactor output control.
- b. Thrust vectoring.
- c. Power allocation.
- d. Fault detection and recovery.

5. *EP subsystem*: This includes ion thrusters, HETs, or VASIMR, selected based on the mission profile. These provide high Isp propulsion suited for long-duration missions.

6. *Heat rejection system*: Large radiator panels dissipate excess heat into space to maintain thermal equilibrium, which is critical for reactor and electronics performance.

7. *Feedback loops*: Bidirectional data flow between the physical subsystems, digital twin, and ANN forming a closed-loop control system and enabling adaptive responses to environmental or internal changes.

IX.C.1. Significance

Emerging AI controllers and digital twin technologies enable predictive diagnostics, real-time optimization, and autonomous fault management. NASA's 2025 experiments on AI-managed NEP systems have shown autonomous trajectory correction and power redistribution during simulated system failures. Digital twins provide a virtual mirror for continuous performance monitoring, essential for multiyear missions where ground intervention is impractical.

IX.C.2. Key Takeaways: NEP for Deep-Space Exploration

The following key takeaways synthesize the technical, historical, and strategic insights discussed throughout this review of nuclear electric propulsion (NEP) systems.

They highlight the performance advantages, system-level considerations, legacy developments, and emerging innovations that collectively define the role of NEP in enabling sustained deep-space exploration and future interplanetary logistics.

1. *Transformative potential*: NEP systems offer high Isp (3000 to 10 000 s) and continuous low-thrust operation, enabling sustained logistics for Mars, the outer planets, and deep-space cargo missions.

2. *System integration*: Compact fission reactors, advanced power conversion units (e.g., Brayton cycles), and electric thrusters, such as ion engines, Hall-effect devices, and MPD systems, form the backbone of NEP architectures.

3. *Historical insights*: Pioneering efforts like SNAP-10A and the Soviet TOPAZ reactors, along with current initiatives (NASA Kilopower, DRACO, ESA FLPP, Roscosmos TEM), illustrate progress and persistent challenges over 6 decades of research.

4. *Key challenges*: Reactor miniaturization, efficient thermal management for megawatt-class systems, lightweight radiation shielding, and internationally coordinated regulatory frameworks are critical areas needing advancement.

5. *Emerging innovations*: AI-enabled autonomy, digital twin technologies, metamaterial-based radiators, and modular reactor architectures are promising pathways to overcome technical bottlenecks.

6. *Comparative insights*: NEP systems surpass NTP and solar sails in payload capacity and mission flexibility for interplanetary missions, but require greater system complexity and higher-cost investments.

7. *Future roadmap*: Focus on demonstrating 100-kW(electric)-class NEP in Earth orbit by 2030, scaling toward megawatt-class reactors, and establishing global policies for safe deployment and end-of-life passivation.

X. CONCLUSION

NEP systems, combining advanced compact fission reactors, high-efficiency electric thrusters, and robust power conversion technologies, are emerging as a transformative solution for deep-space exploration

[1,39]. This review outlined the evolution of NEP architectures, from early experimental systems like SNAP-10A and the Soviet TOPAZ reactors [41,42], to contemporary initiatives, such as NASA's Kilopower and DRACO projects [10,11]. The analysis highlighted core subsystems, integration strategies, and key performance metrics while identifying major engineering challenges, including reactor miniaturization, radiation shielding, and long-term operational reliability [14,62].

The incorporation of AI and digital twin technologies offers significant potential to enhance NEP spacecraft autonomy, enabling predictive maintenance, fault-tolerant operations, and real-time decision making during multiyear interplanetary missions [61].

Looking ahead, future research should emphasize hybrid propulsion configurations (e.g., NEP combined with solar EP or chemical systems) [1,39], modular in-orbit assembly of NEP systems, and AI-augmented mission autonomy. Additionally, establishing standardized international safety protocols and fostering public-private partnerships will be critical for accelerating NEP deployment for both crewed and robotic missions. With these advancements, NEP is poised to transition from conceptual frameworks to becoming a cornerstone of sustainable human and robotic presence beyond Earth orbit.

Acronyms

AI: artificial intelligence

ANN: artificial neural network

ARTES: Advanced Research in Telecommunication Systems

CNSA: China National Space Administration

DARPA: Defense Advanced Research Projects Agency

DC: direct current

DRACO: Demonstration Rocket for Agile Cislunar Operations

DOE: U.S. Department of Energy

DoD: U.S. Department of Defense

EP: electric propulsion

ESA: European Space Agency

FEED: field-emission electric propulsion

FLPP: Future Launchers Preparatory Programme

HET: Hall-effect thruster

Isp: specific impulse
 KRUSTY: Kilowatt Reactor Using Stirling Technology
 MPD: magnetoplasmadynamic
 NASA: U.S. National Aeronautics and Space Administration
 NEP: nuclear electric propulsion
 NEXT: NASA Evolutionary Xenon Thruster
 NSTAR: NASA Solar Technology Application Readiness
 NTP: nuclear thermal propulsion
 PPT: pulsed plasma thruster
 SF: self-field
 SNAP: Systems for Nuclear Auxiliary Power
 TEM: Transport and Energy Module
 TRL: technology readiness level
 VASIMR: Variable Specific Impulse Magneto plasma Rocket

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